

1. Phase 1: Multi-Modal Feature Matrix & Microenvironment Deconvolution

Analytical Methodology & Rationale

- **What is being done:** Loading preprocessed clinical, transcriptomic, and genomic data and constructing dual feature matrices: an ICI-treated cohort ($N_{ICI} = 326$) and a full melanoma cohort ($N_{Full} = 699$) with engineered immune signatures, Macrophage STV ratios, and cell deconvolution metrics.
- **Why we are doing it:** Raw RNA-seq gene expression matrices containing ~19,757 genes suffer from severe dimensionality challenges. Transforming high-dimensional transcriptomics into validated signature scores and cell-type fractions provides interpretable, non-redundant biological features for both response prediction and unsupervised stratification.
- **What question it answers:** How are raw multi-modal datasets harmonised and structured into dual feature matrices to power downstream response-supervised modelling and unsupervised patient stratification?

Phase 1 establishes the foundational dataflow architecture by integrating harmonised multi-modal data across 4 melanoma studies ([Liu 2019](#), [Riaz 2017](#), [Hugo 2016](#), and [TCGA-SKCM](#)). To support distinct analytical requirements across downstream phases, Phase 1 outputs two standardised feature matrices:

Dual Feature Matrix Dataflow Architecture

1. **ICI-Treated Feature Matrix** (`feature_matrix.csv` , $N_{ICI} = 326$):
 - **Composition:** Comprises immunotherapy-treated patients across 4 studies ([Liu 2019](#), [Riaz 2017](#), [Hugo 2016](#), and ICI-treated [TCGA-SKCM](#) subset) with complete RECIST clinical response labels (`RESPONSE_BINARY`).
 - **Downstream Routing:** Powers response-supervised analyses: **Phase 2** (Feature Analysis & Youden Cutoffs), **Phase 5** (Subgroup Predictive Modelling), and **Phase 6** (Clinical Utility & Decision Curve Analysis).
2. **Full Melanoma Feature Matrix** (`feature_matrix_full.csv` , $N_{Full} = 699$):
 - **Composition:** Merges ICI trial cohorts with the complete reference cohort ([TCGA-SKCM](#), $N = 443$), expanding the dataset to capture overall population-level biological heterogeneity.
 - **Downstream Routing:** Powers response-agnostic biological discovery and decision support: **Phase 3** (Unsupervised Patient Stratification & Manifold Projections), **Phase 4** (Phenotype Characterisation & Dynamic Trajectories), and **Phase 7** (3-Arm Decision Support & Treatability Index Scoring).

Biological Feature Engineering & Microenvironment Deconvolution

Rather than evaluating ~19,757 genes independently, Phase 1 projects patient expression profiles onto curated biological axes:

- **Core Immune Signatures:** Tumour Inflammation Signature (`TIS`), Cytolytic Index (`CYT` , mean of `PRF1` and `GZMA`), Interferon-gamma (`IFN_gamma`), `CD8_Tcell` (`CD8A/B` gene average), `CD274` (`PD-L1`) expression, and the Immune Predictive Score (`IMPRES`).
- **Macrophage STV** (`M1_M2_Ratio`): Computed using a linear Signature Transcript Vector (W_g , 14,835 genes) to quantify the microenvironmental balance between pro-inflammatory M1 macrophages ($W_g > 0$) and pro-tumour M2 macrophages ($W_g < 0$).
- **Transcriptomic Cell Deconvolution:** Marker-based signature scores estimating the relative infiltration abundance of CD8+ T cells (`CD8_T_cells`), CD4+ T cells (`CD4_T_cells`), NK cells (`NK_cells`), B cells (`B_cells`), M1 Macrophages (`M1_Macrophages`), M2 Macrophages (`M2_Macrophages`), and Cancer-Associated Fibroblasts (`CAFs`).
- **Engineered Spatial Microenvironment Indicators:** Spatial proxy ratios quantifying cytotoxic T-cell penetration versus stromal exclusion: `Spatial_CD8_CAF_Distance_Ratio` ($\log_2(\text{CD8}/\text{CAF})$) and `Spatial_Tumour_Infiltration_Index` ($\log_2(\text{CD8} \times \text{M1_M2_Ratio}/\text{CAF})$).

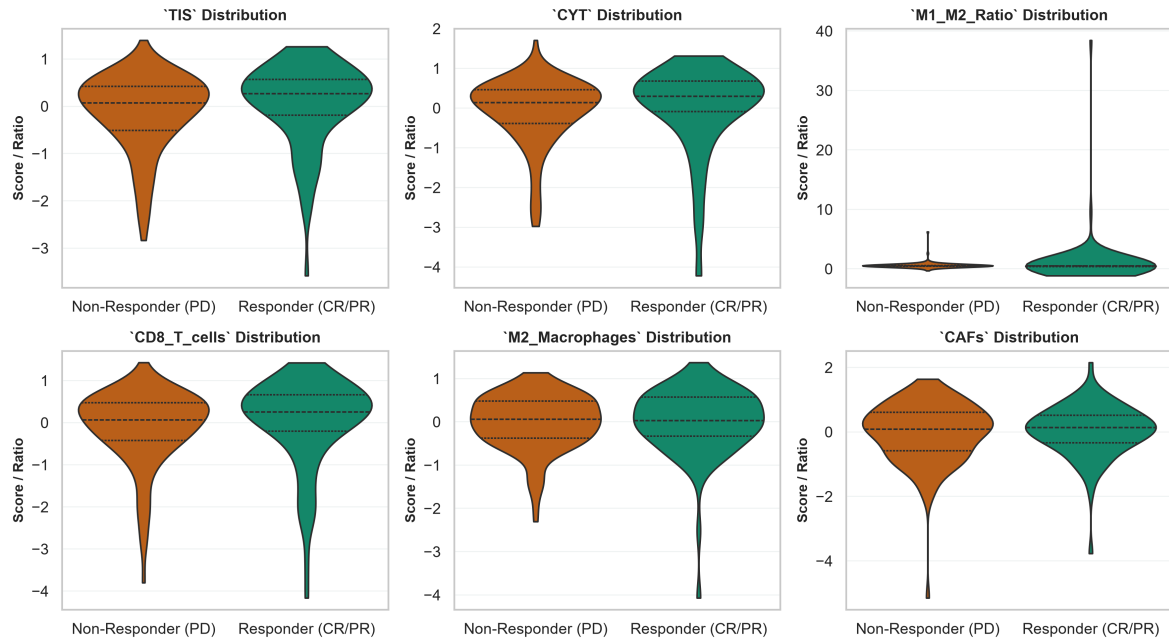
Methodological Scope Note: IMPRES in the Q5 Feature Panel

The Immune Predictive Score (`IMPRES` , [Auslander et al., 2018](#)) evaluates 15 pairwise boolean comparisons between co-stimulatory and co-inhibitory immune checkpoint genes.

- **Scope:** `IMPRES` is **retained** in the 33-feature panel and used as a candidate predictor in **Phase 5** (Subgroup Predictive Modelling) and **Phase 6** (Clinical Utility Analysis). It is **not** part of the TME deconvolution core (Macrophage STV, cell-type fractions, or spatial proxy indicators) because it encodes a discrete checkpoint pairwise logic rather than a continuous microenvironment abundance estimate.
- **Known Limitation:** Key co-stimulatory partners (`CD28` , `CD86` , `CD80` , `CD40` , `CD200` , `TNFRSF4` , `VSIR`) are absent or unmapped in a subset of merged multi-study expression matrices. Where co-stimulatory genes are missing, `IMPRES` is computed over a reduced pair set — producing a score that may underestimate true checkpoint activity for those samples.
- **Collinearity:** `IMPRES` exhibits moderate collinearity with continuous T-cell signatures (`TIS` , `CYT` , $r_s > 0.75$). Phase 5 regularisation (Random Forest, LOCO CV) mitigates this.

Baseline Biomarker Feature Distributions

Baseline Immune & Microenvironmental Biomarker Distributions (N=195)



📌 Key Takeaways

- **Dual-Matrix Dataflow:** Established a dual dataflow pipeline isolating response-labeled ICI trials ($N_{ICI} = 326$) for predictive modelling while embedding the full cohort ($N_{Full} = 699$) for unsupervised manifold learning.
- **Dimensionality Reduction:** Compressed ~19,757 transcriptomic features into 19 engineered biological signatures (part of a 33-feature multi-modal panel, centered on 9 core baseline biomarkers).
- **M1/M2 Polarisation:** The Macrophage STV score captures stromal microenvironmental suppression that operates independently of total T-cell density.

Phase 1 Feature Matrix Architecture & Complete Feature Inventory

- **Transcriptomic Features (19):**
 - **Core Immune Signatures (6):**
 1. TIS (Tumour Inflammation Signature)
 2. CYT (Cytolytic Index)
 3. IFN_gamma (Interferon-gamma signalling)
 4. CD8_Tcell (*CD8A/B* gene average)
 5. PD_L1 (*CD274*).
 6. IMPRES (Immune Predictive Score — retained for Phase 5/6 predictive modelling).
 - **Macrophage STV Metrics (4):**
 1. M1_score
 2. M2_score
 3. M1_M2_Ratio
 4. Macrophage_STV_Score (Signature Transcript Vector balance).
 - **Transcriptomic Cell Deconvolution (7):**
 1. CD8_T_cells
 2. CD4_T_cells
 3. NK_cells
 4. B_cells
 5. M1_Macrophages
 6. M2_Macrophages
 7. CAFs (Cancer-Associated Fibroblasts).
 - **Spatial Microenvironment Indicators (2):**
 1. Spatial_CD8_CAF_Distance_Ratio (log2 CD8 / CAF proxy ratio)
 2. Spatial_Tumour_Infiltration_Index (log2 CD8 x M1_M2_Ratio / CAF index).
- **14 Genomic, TMB, Neoantigen & Genomic Instability Features:**
 - **Driver Mutations (3):**
 1. mut_BRAF
 2. mut_NRAS
 3. mut_NF1 (binary oncogenic driver status).
 - **TMB, Neoantigen Burden & Genomic Instability (11):**
 1. TMB_NONSYNONYMOUS
 2. SNV_NEOANTIGEN
 3. INDEL_NEOANTIGEN
 4. FUSION_NEOANTIGEN
 5. SPLICE_NEOANTIGEN
 6. CTA_SELF_NEOANTIGEN
 7. VIRUS_NEOANTIGEN
 8. ERV_NEOANTIGEN
 9. ANEUPLOIDY_SCORE
 10. MSI_SCORE_MANTIS
 11. MSI_SENSOR_SCORE .
- **9 Core Baseline Biomarkers (Primary Subset):**
 1. TIS
 2. CYT
 3. PD_L1
 4. Macrophage_STV_Score
 5. CD8_T_cells
 6. CD4_T_cells
 7. NK_cells
 8. B_cells
 9. CAFs (used for primary volcano, Youden ROC, and radar visualisations).

Phase 1 Script Execution & Software Module Architecture

- **Primary Pipeline Execution Scripts:**
 - **01_load_and_prepare.py**: Loads preprocessed clinical, expression, and genomic data from `data/processed/merged/`, computes immune signatures (`TIS`, `CYT`, `IFN_gamma`, `CD8_Tcell`), calculates Macrophage STV (`M1_M2_Ratio`), runs transcriptomic cell deconvolution, and exports dual feature matrices (`feature_matrix.csv`, $N_{ICI} = 326$; `feature_matrix_full.csv`, $N_{Full} = 699$).
- **Core Supporting Python Modules:**
 - **deconvolution.py**: Implements Signature Transcript Vector (`compute_macrophage_stv`) and cell-type abundance estimation (`compute_cell_deconvolution`).
 - **phenotyping.py**: Renders 2x3 faceted biomarker distribution violins (`plot_baseline_signature_boxplots`).
 - **q5_constants.py**: Single source of truth for cell marker panels (`CELL_TYPE_MARKERS`), immune signature definitions (`IMMUNE_SIGNATURE_MARKERS`), and neutral STV fallback ratios.
- **Shared Cross-Question & Pipeline Modules:**
 - **run_q5_pipeline.py**: Master pipeline orchestrator executing `01_load_and_prepare.py` as Step 1.
 - **generate_q5_report.py**: Compiles live statistical summaries and generates phase markdown reports.